TASK - 2

Project : Automatic LED ON/OFF Using LDR Sensor in Tinkercad

AIM:

To stimulate an IOT based automated lighting system using an LDR sensor an Arduino in Tinkercad.

Components Required:

* Arduino UNO
* LDR (Light Dependent Resistor)
* LED
* 220Ω Resistor
* Breadboard
* Jumper Wires

Principle:

* The LDR sensor detects the surrounding light intensity. When the environment becomes dark, the resistance of the LDR changes, and the Arduino turns ON the LED automatically. When there is sufficient light, the LED turns off.

This project demonstrates a basic smart lighting system used in homes and street lights.

Circuit Connections:

LDR Connection

* One terminal → 5V
* Other terminal → A0 pin and resistor to GND

LED Connection:

* Positive terminal → Digital Pin 13
* Negative terminal → GND through 220Ω resistor

Applications:

* Smart street lights
* Automatic room lighting
* Energy-saving systems
* Security lighting systems

Arduino Code:

```
#include <iostream>
using namespace std;

int main()

    int ldr = A0;
    int led = 13;
    int value = 0;

void setup()
{
    pinMode (led,OUTPUT);
    Serial.begin (9600);
}

void loop()
{
    value = analogRead (ldr);
    Serial.println (value);

    if (value < 500)
    {
        digitalWrite (led, HIGH);
    }
```

```
else
  {
      digitalWrite (led, LOW);
  }
    delay (500);
}
```

Code Explanation:

* analogRead (ldr) reads the light intensity value.

* If the value is less than 500, it indicates darkness.

* The LED turns ON automatically.

* If the value is greater than 500, the LED turns OFF.

* Serial.println (value) displays sensor values in the serial monitor.

Output:

In Bright Light

* LDR detects high light intensity
* LED remains OFF

In Darkness

* LDR detects low light intensity
* LED remains and turns ON automatically

Limitations:

* Accuracy depends on sensor quality
* LDR performance may chance over time.
* External light sources can affect readings.
* Limited range for detecting light intensity

Future Enhancements:

This project can be improved further by:

* Connecting to IOT cloud platforms
* Using Wi-Fi modules like ESP8266.
* Monitoring light intensity remotely using mobile apps.
* Using solar-powered systems for energy efficiency.

Real Life Applications of IOT Smart Lighting :

Many modern cities use IOT - based smart lighting systems to save electricity and improve efficiency.

Examples include:

* Automatic highway lights
* Smart office lighting
* Smart parking systems
* Intelligent building management system

These systems reduce electricity consumption and maintenance costs.

Circuit design Bodacious Jofo-D

tinkercad.com/things/9umNJjrZkBA/editel?returnTo=%2Fcircuits

TIN KER CAD

Bodacious Jofo-Densor

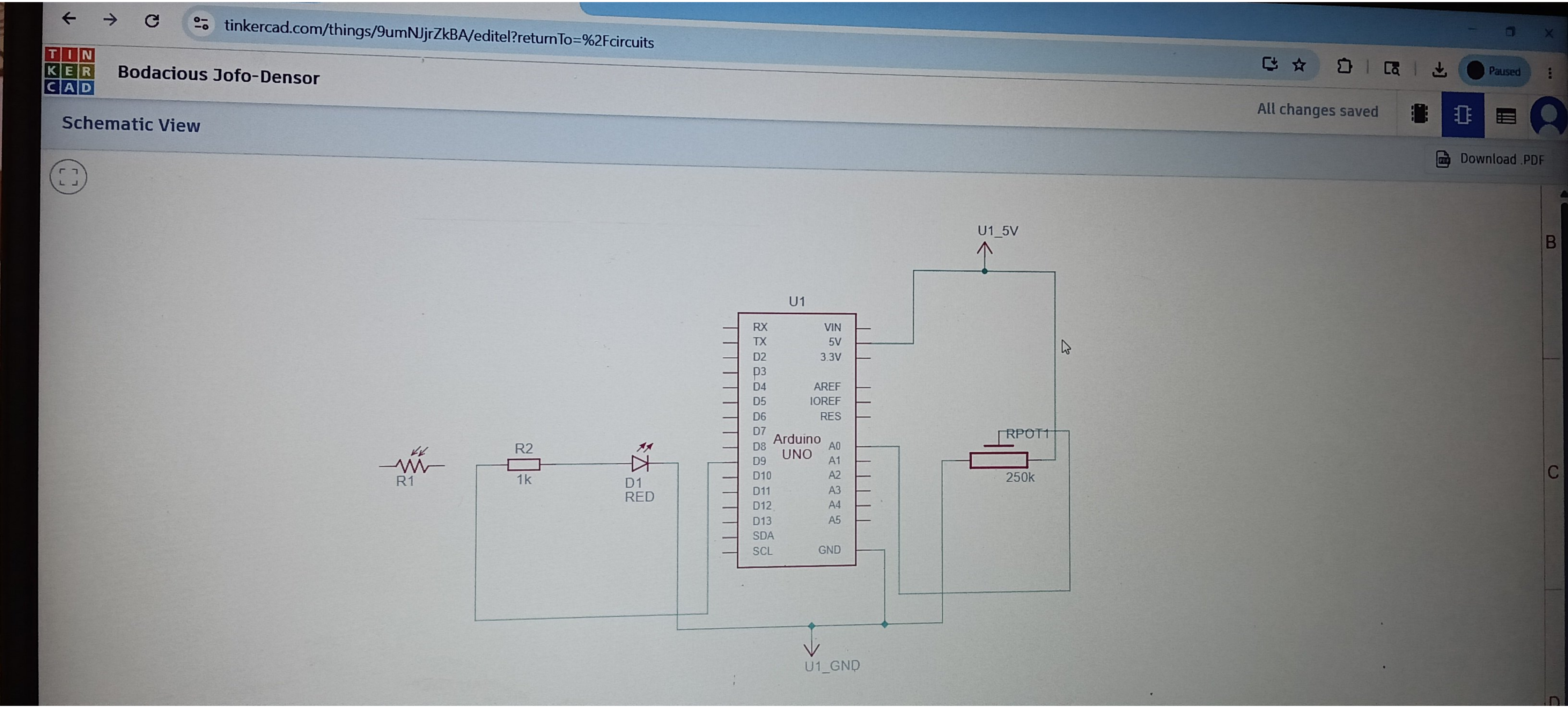
Paused

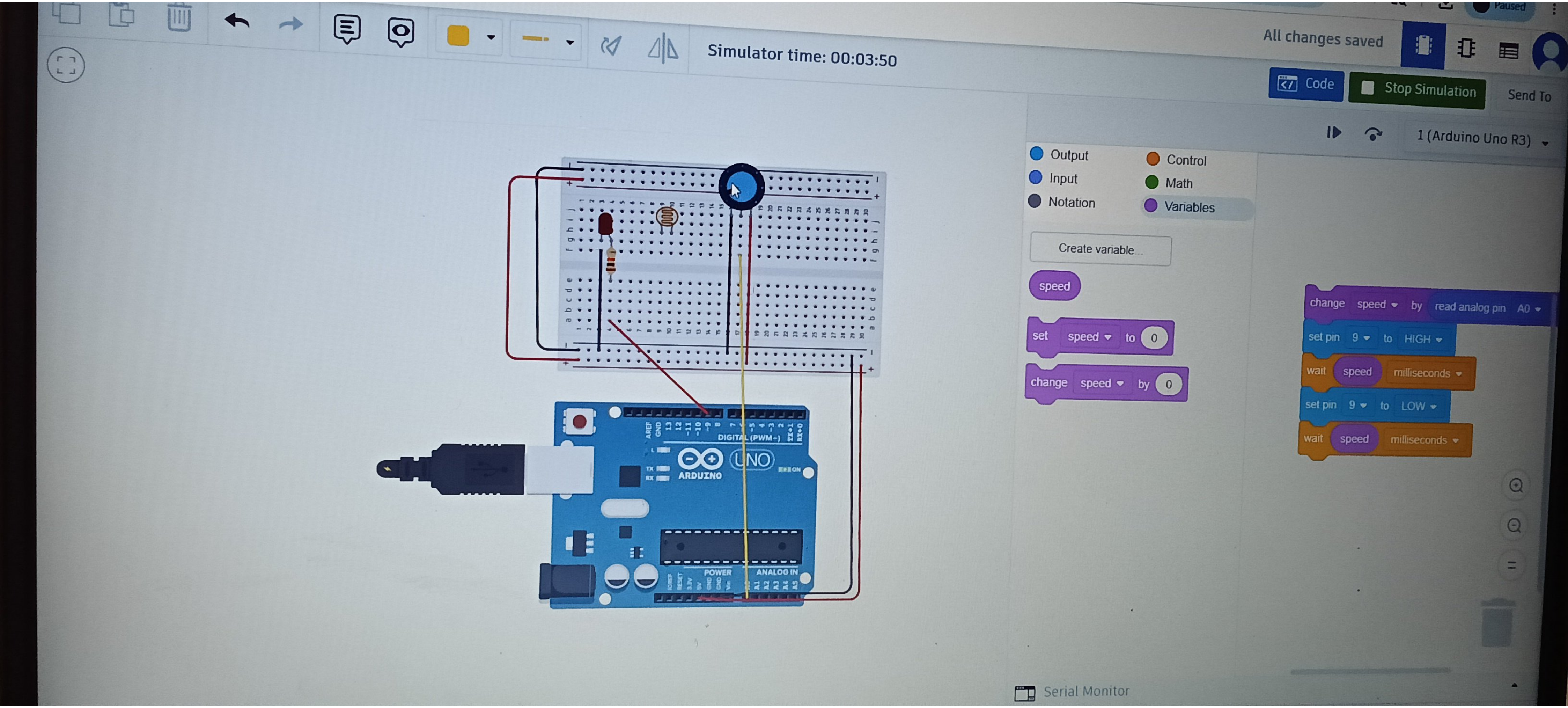
Component List

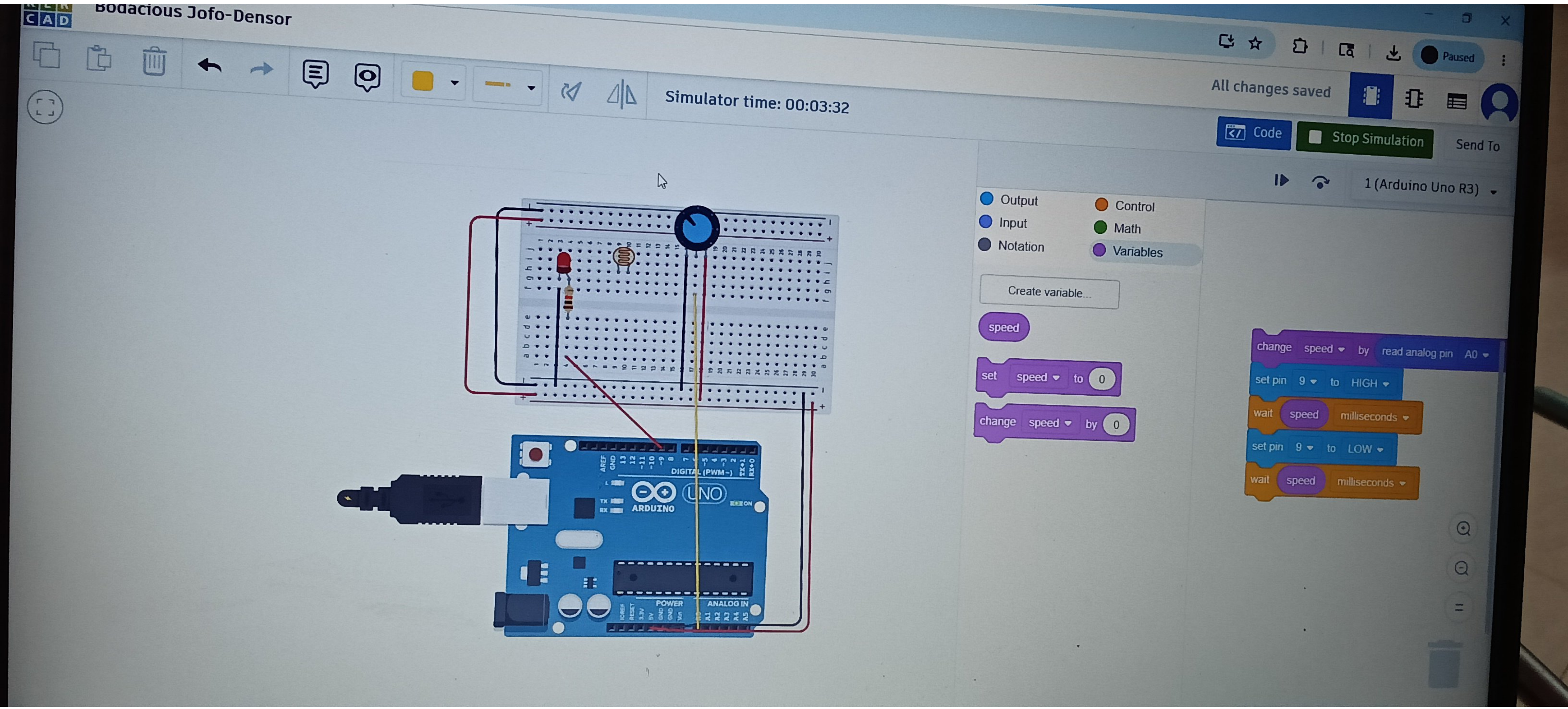
All changes saved

Download CSV

Name	Quantity	Component
U1	1	Arduino Uno R3
R1	1	Photoresistor
D1	1	Red LED
R2	1	1 kΩ Resistor
Rpot1	1	250 kΩ Potentiometer


tinkercad.com/things/9umNJjrZkBA/editel?returnTo=%2Fcircuits
TIN
KER
CAD
Bodacious Jofo-Densor
Paused
All changes saved
Schematic View
Download .PDF
U1_5V
U1
RX
TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
SDA
SCL
VIN
5V
3.3V
AREF
IOREF
RES
Arduino
UNO
A0
A1
A2
A3
A4
A5
GND
R1
R2
1k
D1
RED
RPOT1
250k
U1_GND
B
C


Simulator time: 00:03:50
All changes saved
Code
Stop Simulation
Send To
1 (Arduino Uno R3)
Output
Input
Notation
Control
Math
Variables
Create variable...
speed
set speed to 0
change speed by 0
change speed by read analog pin A0
set pin 9 to HIGH
wait speed milliseconds
set pin 9 to LOW
wait speed milliseconds
DIGITAL (PWM~)
ARDUINO
UNO
POWER
ANALOG IN
Serial Monitor


Bodacious Jofo-Densor
Simulator time: 00:03:32
All changes saved
Code
Stop Simulation
Send To
1 (Arduino Uno R3)
Output
Input
Notation
Control
Math
Variables
Create variable...
speed
set speed to 0
change speed by 0
change speed by read analog pin A0
set pin 9 to HIGH
wait speed milliseconds
set pin 9 to LOW
wait speed milliseconds
ARDUINO
UNO
DIGITAL (PWM~)
POWER
ANALOG IN